Lesson notes · for you

Exact instructions

Programming · Level 1 · Lesson 1

Time	Materials
20 min	Paper and a pencil; Paper cards with arrows; A phone (optional)

Learning goals

- Say what an algorithm is.
- Give exact instructions.
- Program the robot with cards.

Activity: The sandwich robot

1. Warm-up (3 min): Imagine telling a robot “make a sandwich”. It doesn’t move. What does it need from you?
2. Write (7 min): On paper, write every step of making a jam sandwich, in order. Number the steps.
3. Test (7 min): Swap with a partner. Read their steps aloud while they act them out exactly, with no guessing. Laugh at what goes wrong!
4. Fix (3 min): Add the missing steps, or swap the ones in the wrong order, and test again.

Act it out with pretend bread: no food is needed.

Discussion

- What happened when a step was missing?
- Why can’t a robot guess?
- Where else do you follow steps in order?

Check yourself

- Write three exact steps for one thing you do every day.

Try it this week

Write the steps for making a jam sandwich, or a cup of tea, for a robot. Ask someone at home to follow your steps exactly, without guessing. Did anything go wrong?

Worksheet

Exact instructions

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Name _____ Class _____ Date _____

1 **"Tidy your desk." Who understands that, with no more steps?**

◯ A friend ◯ A robot ◯ Both of them

2 **Put the steps for making a jam sandwich in order.**

☐ Spread jam on one slice

☐ Take two slices of bread

☐ Cut the sandwich in half

☐ Put the other slice on top

3 **Help the robot reach the flag. Tap a card in your program to repeat it.**

→			★

Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

4 **If you forget a step, the robot guesses what you meant.**

◯ True ◯ False

5 **Now the flag is around a corner. Build a program that gets there.**

		★	
↑			

Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

✎ **Think of a job you do every morning. Write its steps in order. Which step would a robot get wrong?**

_____

_____

_____

Check your answers

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1 **A friend**

A friend can guess what to do. A robot needs every step: pick up the book, put it on the shelf, put the pens in the box...

2 **Top to bottom: 2, 1, 4, 3**

You need the bread before the jam, and the jam before the top slice.

3 **Any program that takes the robot to the ★ with 2 cards or fewer.**

The flag is straight ahead, three squares away: one card, forward, repeated 3 times.

4 **False**

A robot never guesses. It does exactly what the cards say, so every step must be there.

5 **Any program that takes the robot to the ★ with 4 cards or fewer.**

Forward 3 times, turn right, then forward 2 times: three cards.